

Varshith Tipirneni

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EDUCATION

University of North Carolina at Chapel Hill (GPA - 4.0/4.0)

Aug 2024 – May 2026

Masters in Statistics, Analytics and Data Science

Chapel Hill, NC

Coursework: Machine Learning, Deep Learning, Time Series Analysis, Statistical Modeling, Stochastic Models

EXPERIENCE

Data Analyst | Viatris | Remote

May 2025 – May 2026

- Resolved supply chain tracking gaps by mapping daily data workflows with supply planners to scope system technical requirements and identify operational bottlenecks.
- Built SQL data infrastructure to centralize distributed procurement datasets into indexed database tables handling over 100K rows.
- Configured automated Alteryx ETL workflows to remove inactive PO lines, decreasing active dashboard data overhead by 88%.
- Developed a data transformation pipeline to compute predictive Out-of-Stock parameters, enabling planners to intercept high-risk supply gaps and lower total gap cost by 25%.

Solutions Architect (Data Science Practicum) | NMSS | Remote | [Website](#)

Aug 2025 – May 2026

- Designed a multi-stage sequential execution system to model longitudinal patient pathways across 1M medical claims records.
- Built multi-table relational SQL query layers to extract disparate client histories into a consolidated, analytics-ready database.
- Trained and deployed tree-based classification pipelines, achieving a 0.95 overall AUC score in predicting clinical medication switches.
- Validated pipeline performance on held-out test frames, outperforming legacy flat-table architectures by a 5.0% absolute accuracy margin.

Catalog Solutions Associate | Knowde | Bangalore, India

Mar 2024 – Jun 2024

- Developed automated document parsing scripts to extract product specifications and chemical dimensions from over 10K unstructured technical data sheets.
- Audited cross-database supplier records to reconcile system structural inconsistencies and maintain data integrity for downstream search routing.
- Integrated local LLM context scripts to normalize technical catalog descriptions, improving term indexing accuracy and search discovery rates.

Technical Services Intern | Reliance Industries Limited | Jamnagar, India

Sept 2022 – Nov 2022

- Executed crude distillation unit operational simulations using Petrosim to analyze process boundary variables and identify efficiency changes.
- Built parameterized analytical models in Excel to evaluate fired heater thermal profiles, generating functional performance benchmarks for plant operations.

PROJECTS

PULSE – Agentic Real-Time Analytics Pipeline

May 2026 – Present

- Architected a concurrent multi-agent collection system using Python thread pools to coordinate asynchronous web, sentiment, and trend mining scripts.
- Built a FastAPI backend utilizing Server-Sent Events (SSE) and worker queues to process over 5K token streams per minute directly to a live client dashboard.

HybridRAG – Enterprise Pharma Compliance Platform | [Website](#)

Mar 2025 – Present

- Developed a 6-label compliance engine pairing BGE dense vector embeddings via pgvector/HNSW indexing with BM25 lexical search through Reciprocal Rank Fusion.
- Implemented explicit exception logic to route low-confidence outputs (< 70%) into human-in-the-loop validation queues backed by a GitHub Actions collection runner.

TECHNICAL SKILLS

Solutions Engineering: Solutions Architecture, Requirements Scoping, Technical Product Demos, Technical Workflow Mapping, Human-in-the-Loop Architecture

Languages & Frameworks: Python, SQL, FastAPI

Data Architecture & Infrastructure: Pipeline Engineering, ETL/ELT Systems, PostgreSQL, Snowflake, Vector Databases (pgvector, Qdrant), Alteryx, Power BI

AI/ML Deployment: Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration, MLOps, LLMOps, Model Evaluation Frameworks

Cloud & Systems: Kubernetes, Docker, AWS (EC2/S3), GitHub Actions, CI/CD pipelines, Git